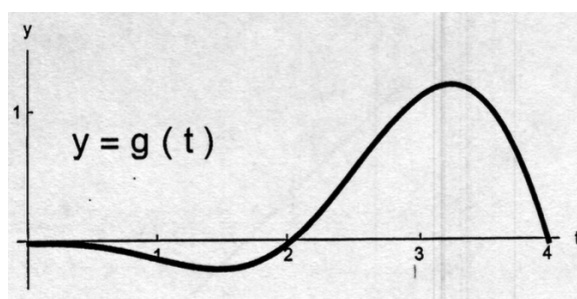


Problem 2

Problem 2

Problem 1 The graph of g , a continuous function on $[0, 4]$, is shown in the figure. Let $A(x) = \int_0^x g(t) dt$, for $0 \leq x \leq 4$.



- (a) Circle the correct statement about $A(2)$.
- (i) $A(2) = 0$;
 - (ii) $A(2) > 0$;
 - (iii) $A(2) < 0$;
 - (iv) none of the previous answers.
- (b) Circle the correct statement about $A(3.8)$.
- (i) $A(3.8) = 0$;
 - (ii) $A(3.8) > 0$;
 - (iii) $A(3.8) < 0$;
 - (iv) none of the previous answers.
- (c) Circle the correct statement about $A'(3.8)$.
- (i) $A'(3.8) = 0$;
 - (ii) $A'(3.8) > 0$;
 - (iii) $A'(3.8) < 0$;
 - (iv) none of the previous answers.
- (d) Find the solution of the following initial value problem: $y'(x) = g(x)$ and $y(0) = 2$.

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- (i) $y(x) = g(x)$;
- (ii) $y(x) = g(x) + 2$;
- (iii) $y(x) = A(x)$;
- (iv) $y(x) = A(x) + 2$;
- (v) $y(x) = g'(x)$;
- (vi) $y(x) = g'(x) + 2$;
- (vii) *none of the previous answers.*

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- (e) Find the expression for $\int_0^4 |g(t)| dt$.

- (i) $A(4)$;
- (ii) $A(2) - A(4)$;
- (iii) $A(4) - A(2)$;
- (iv) $A(4) - 2A(2)$;
- (v) *none of the previous answers.*

- (f) Find the midpoint Riemann sum for the function g on the interval $[0, 4]$ with $n = 2$ (the number of subintervals).

- (i) $g(1) + g(3)$;
- (ii) $g(0) + g(2)$;
- (iii) $g(2) + g(4)$;
- (iv) $2g(1) + 2g(3)$;
- (v) $2g(0) + 2g(2)$;
- (vi) $2g(2) + 2g(4)$;
- (vii) *none of the previous answers.*